INCH-POUND
MIL-H-965E(SH)
22 December 1994
SUPERSEDING
MIL-H-965D(SHIPS)
29 November 1967
(See 6.14)

MILITARY SPECIFICATION

HEATERS, WATER, ELECTRIC, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command (NAVSEA), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes electric, self contained, water heaters as a complete system and ensures that all system components are compatible and properly integrated with regard to size, function, and performance. The heater systems must comply with system integration requirements (see 6.9), performance, endurance, safety test procedures, shock and vibration, and standardization for installation on U.S. Naval surface ships and submarines.

1.2 Classification. The class and size of the heaters specified (see 6.2) are identified by the alpha/numeric designation in table I which indicates its application characteristics.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Class and size identification.

Class, size designation	Class	Tank capacity (gallons)		
		A	B	C
A 5		5		
A 10		10		
A 25		25		
A 50		50		
B 5			5	
B 10			10	
B 25			25	
B 50			50	
C 5				5
C 10				10
C 25				25
C 50				50

Classes

Class A - Personnel use; 125°F (example: showers and lavatories).
Class B - Non personnel use; 160°F (example: grease interception hoods and pot sanitizing sinks.
Class C - Non personnel use; booster heater; 180°F (example: laundry and scullery).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact): Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-R-2033	- Relays - Power, Electrical Specialty, Induction Type, Naval Shipboard Use.
MIL-E-2036	- Enclosures for Electric and Electronic Equipment.
MIL-C-2212	- Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices.
MIL-S-4040	- Solenoid, Electrical General Specification for.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
DOD-P-15328	- Primer (Wash), Pretreatment (Formula No. 117 for Metals). (Metric)
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-W-16878	- Wire, Electrical, Insulated, General Specification for.
MIL-W-16878/1	- Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 600 Volts.
MIL-W-16878/2	- Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 1000 Volts.
MIL-W-16878/3	- Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 3000 Volts.
MIL-W-16878/17	- Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 600 Volts, Polyamide Jacket.
MIL-W-16878/18	- Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 1000 Volts, Polyamide Jacket.
MIL-W-16878/19	- Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105°C, 3000 Volts, Polyamide Jacket.
MIL-I-17244	- Indicators, Temperature, Direct-Reading, Bimetallic (3-Inch and 5-Inch Dial).
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts), Packaging of.
MIL-G-18997	- Gauge, Pressure, Dial Indicating.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-F-20042	- Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-H-22577	- Heating Elements, Electrical: Cartridge, Strip and Tubular Type.
MIL-T-24270	- Thermowells for Thermometers and Electrical Temperature Sensors, General Specification for.
MIL-T-24270/1	- Thermowells for Thermometers and Electrical Temperature Sensors; Insertion Length - 2 Inches, Connection-Socket Weld or Socket Brazed, 3/4-Inch NPS.
MIL-G-24696	- Gasket, Sheet, Non-Asbestos.

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STANDARDS

MILITARY

MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-248	- Welding and Brazing Procedure and Performance Qualification.
MIL-STD-278	- Welding and Casting Standard.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-769	- Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-777	- Schedule of Piping, Valves, Fittings and Associated Piping Components for Naval Surface Ships.
MIL-STD-781	- Reliability Testing for Engineering Development, Qualification, and Production.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1399 Section 300	- Interface Standard for Shipboard Systems Electric Power, Alternating Current. (Metric)
MIL-STD-1686	- Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices). (Metric)
DOD-STD-2169	- High Altitude Electromagnetic Pulse Environment.
MIL-STD-45662	- Calibration Systems Requirements.

HANDBOOK

MILITARY

DOD-HDBK-263	- Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Devices). (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-001-7000 - Piping Systems, Brazed, Fabrication and Inspection.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Z 21.22 - Standard for Relief Valves and Automatic Gas Shutoff Devices for Hot Water Supply Systems. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
B 16.5 - Pipe Flanges and Flanged Fittings. (DOD adopted)
BPVC Section IV - Rules for Construction of Heating Boilers.
BPVC Section VIII - Rules for Construction of Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 276 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes.
B 96 - Standard Specification for Copper-Silicon Alloy Plate, Sheet, Strip, and Rolled Bar for General Purposes and Pressure Vessels.
B 171 - Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers.
D 4727 - Standard Specification for Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.6) in accordance with 4.4.

3.2 Materials. Unless otherwise specified (see 6.2 and 6.22), materials used in manufacturing the heaters shall be as listed in table II. Material compatibility shall be in accordance with MIL-STD-889.

TABLE II. Electric hot water heater system material list.

Part	Material
Element flange; Tank shell; heads or cover plates	90-10 copper-nickel (ASTM B 171) Copper-silicon (ASTM B 96)
Tank flanges; Element support; all parts in with water	90-10 copper-nickel (ASTM B 171) Copper-silicon (ASTM B 96)
Gaskets	MIL-G-24696
Jacket; fasteners	Corrosion-resistant metal or materials suitably protected against corrosion (see 3.2.1)
Insulation	Fiberglass wool (MIL-STD-769)

3.2.1 Corrosion protection.

3.2.1.1 Corrosion-resisting materials. Corrosion-resisting steel, copper, bronze, copper-silicon, copper-nickel, copper-nickel alloys, or other specified materials (see 6.11) are considered satisfactory corrosion-resisting materials. Where corrosion-resistant steel is used, it shall be type 304 or 316 in accordance with ASTM A 276. Corrosion-resistant steel, when fabricated by any method that tends to reduce corrosion resisting properties, shall be normalized to restore those properties before being used in the assembly of any unit.

3.2.1.2 Noncorrosion resisting materials. Parts other than corrosion-resistant materials shall be protected against corrosion by the use of chemicals, electrolytic processes, plating, or paints and enamels.

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3.2.1.3 Hardware. Bolts, nuts, studs, pins, screws, and other such fastenings or fittings shall be of a corrosion-resistant material. Self-tapping screws with machine screw threads may be used in the cabinet assembly. Sheet metal screws with sheet metal threads shall not be permitted.

3.3 Construction. Heaters shall be constructed as specified in 3.1 through 3.3.3 (see 6.3 and 6.8).

3.3.1 General. The electric water heater system shall be a complete system in which all components and materials are compatible and properly integrated with regard to size, function, and performance. The heater shall heat or discharge water at the recommended temperature and capacity upon demand. A natural circulation system which entails the use of shrouds shall be employed to minimize water stratification. Heater systems shall have good accessibility to all internal and external parts for ease of serviceability. This requirement shall prohibit the location of inlet and outlet lines and auxiliary components from obstructing flange connections and other field removable parts. In addition, the water heater shall be constructed so that the tank can be cleaned and the heating elements and thermostatic switch easily removed without disturbing the insulation on the tank or removing any piping. Auxiliary water heater system components (pressure relief valve, heating elements, thermometer, drain connection, drain valve, and magnetic contactor) shall be installed to provide accessibility for ease of (1) inspection; (2) tool maneuverability, and (3) removal or repair of these components. See figure 1 for typical electric water heater systems. The equipment shall reflect good serviceability as specified in 7.13.

3.3.2 Manpower and personnel. The water heater system shall be constructed to be operated and maintained by two seamen with training equivalent to a competent journeyman with skills in plumbing and electrical.

3.3.3 Response time. The time required for the heater to generate hot water (to 130 degrees Fahrenheit (°F)) when the tank is filled with water at 40°F (see 3.5.2(a)), shall be not greater than 1 hour.

3.4 Reliability requirements. Each heater system shall have an operating life of 100,000 hours (see 6.3). The water heater systems shall have a Mean Time Between Failures (MTBF) of 10,000 hours of operation, Mean Time to Repair (MTTR) of 12 hours, Mean Time Between Corrective Maintenance (MTBCM) of 1,000 hours and an Operational Availability (A_o) of 0.95. The MTBF shall be supported by analysis or industry information provided and verified by the producer.

3.4.1 System failure. More than 2 hours of corrective maintenance required on the heater system shall be considered a failure.

3.5 Voltages. The heater system shall operate on either of the following power supplies or as specified (see 6.2).

(a) 440 volts (V) (nominal), 3 phase (ph), 60 hertz (Hz), alternating current (ac) (see 3.5.2(c)).
(b) 115 or 230 V (nominal), direct current (dc) (see 3.5.2(c)).

Note: See 3.7.10.9 (Electrical equipment) for other voltages.

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3.5.1 Performance conditions. Heater systems shall meet the specified performance requirements under the following parameters, unless otherwise specified:

(a) Heater: Inclination: (see 3.7.8).
(b) Inlet water temperature: 40°F. (Unless otherwise specified, see 6.2.)
(c) Inlet water pressure: 30 to 110 pounds per square inch (lb/in^2) gauge, unless otherwise specified (see 6.2).
(d) Inlet water temperature: (see 1.2 and 3.6).

3.5.2 Minimum and maximum design conditions. The heater systems shall meet all specified performance requirements under the following withstanding conditions unless otherwise specified (see 6.2):

(a) Inlet water temperature: (40 to 90°F).
(b) Inlet water pressure: 30 to 100 lb/in^2 gauge.
(c) Voltage ranges: All combatants: 400 volts alternating current (Vac) + 5 percent.
Submarines: 180 to 355 Vdc.
(d) Outlet water temperatures:

Class	Min	Max
A	125°F	140°F
B	155°F	165°F
C	175°F	185°F

3.6 Temperature adjustments. Heaters shall be adjustable between the following temperature ranges: class A, 90 and 140°F; class B, 90 and 160°F; and class C, 130 and 180°F. Class C electric water heaters are booster heaters and shall only be used in conjunction with class A or class B electric water heaters (see 3.7.10.1).

3.7 Fabrication. The water heater system shall meet the requirements of ASME BPVC Section VIII and MIL-STD-278 and MIL-STD-248 for class A-4 pressure vessels (see 3.7.6). The shell shall maintain a working pressure of not less than 100 lb/in^2 gauge unless otherwise specified (see 6.2). The water heater shall consist of a pressure tank which shall be insulated against heat loss and protected by a jacket.

3.7.1 Tanks. Tanks shall consist of a shell with heads or cover plates that shall be welded or silver brazed as specified in 3.7.6. The tank shall be furnished with a base support that is either welded, silver brazed, or bolted to the tank. The base support shall be deck mounted by a minimum of four bolts measuring 5/8-inch, 11 UNC.

3.7.1.1 Shells. Shells shall be cylindrical, fabricated from sheet material, as specified in table II, and either silver brazed or seam welded in accordance with 3.7. A means for internally supporting the heating element(s) during field removal and reinsertion shall be provided. A support that secures the heating element(s) in both horizontal and vertical directions shall be provided opposite to the heating element mounting flange(s).

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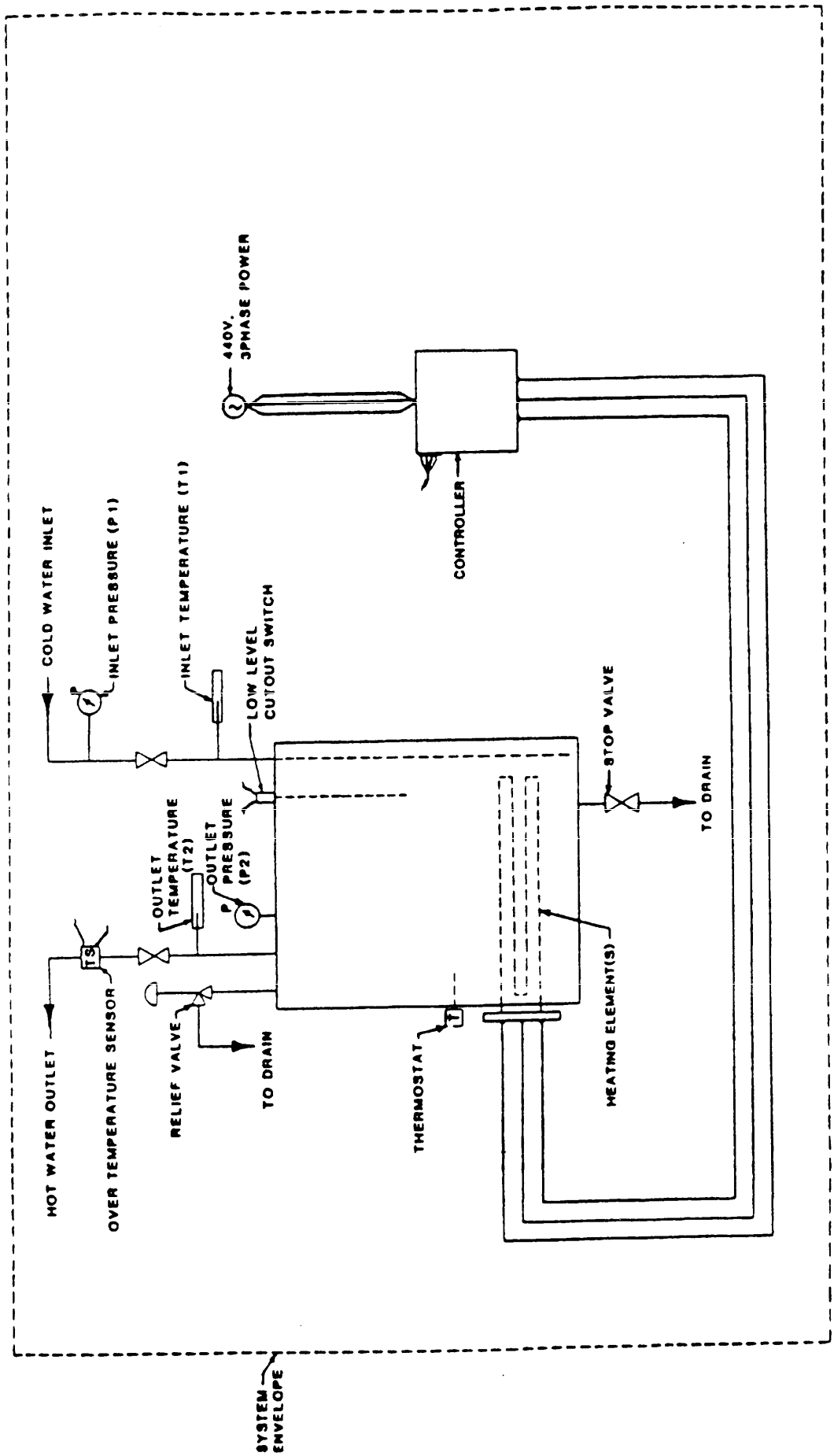


FIGURE 1. Typical electric water heater system.

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3.7.1.2 Heads or cover plates. Heads or cover plates shall be either cast or forged and shall be removable by bolting at the trunk shell flange. The head or cover plate material shall be as specified in table II.

3.7.1.3 Relief valve bosses. Heaters shall be equipped with a boss for a relief valve. Connections for the relief valve's boss shall be a straight threaded union in accordance with MIL-STD-777 (see 3.7.10.3). The boss shall be ground flat to accommodate an O-ring to form a tight seal.

3.7.2 Jacket. Jackets shall be used to enclose the tank and insulating material. The jacket material shall be as specified in table II.

3.7.3 Flanges. Flanges shall be welded to the tank and shall be class 150 in accordance with ANSI B 16.9 or MIL-F-20042.

3.7.4 Heating elements. Heating elements shall be of the type III, tubular type, conforming to MIL-H-22577. The heating elements shall be secured to the element flanges by brazing. The element flanges shall be of the same material as the shell and shall have CRES-sheathed heating elements (see 3.7.1.1).

3.7.5 End connections. End connections for the water inlet and outlet shall terminate in take down connections. The water inlet and outlet shall be supplied with flanges in accordance with ANSI B 16.5, class 150 or MIL-F-20042. Threaded pipe connections shall not be used. Heaters shall be equipped with necessary drain valves to permit complete drainage of the heater system.

3.7.6 Welding and silver brazing. Welding shall be in accordance with MIL-STD-278. In addition, qualification of welders and brazers shall be in accordance with MIL-STD-278 and 0900-LP-001-7000.

3.7.7 Heater supports. The heater systems shall include a frame to support and elevate all heater components to a height permitting all connections to be easily accessible without having to remove framework or any other obstruction (see 3.7.1).

3.7.8 Heater inclination. Water heaters shall operate properly under permanent 15-degree inclination in any direction and symmetrical rolling to 45 degree inclination in any direction.

3.7.9 Identification and information plates. Heaters shall have an identification plate mounted on the terminal box or on the tank in full view, and it shall contain the following data:

(a) ELECTRIC WATER HEATER (in capital letters).
(b) Manufacturer's name and address.
(c) Manufacturer's model, serial, and part numbers.
(d) Class and size.
(e) Contract or purchase order number.
(f) National stock number.
(g) Voltage, wattage, phases, and amperage.
(h) Design outlet temperature.
(i) Maximum outlet temperature.

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Identification plates shall be in accordance with type C of MIL-P-15024 and MIL-P-15024/5. Each controller shall include a schematic diagram in accordance with MIL-C-2212. The diagram shall be secured inside the controller in a location that is readily accessible.

3.7.10 Controls, safety devices, and indicators. Controls, safety devices, and indicators, as shown on figure 1 shall be furnished as part of the water heater system.

3.7.10.1 Thermostatic switches. Thermostatic switches shall be the direct connected immersion bulb type (without capillary tubing). Unless otherwise specified (see 6.2), thermostatic switches shall be adjustable between the temperature ranges specified in 3.6. Thermostatic switches shall maintain the outlet temperature to plus or minus 5°F of the set temperature. The maximum temperature of the water at the top of the tank shall not be greater than the actual thermostat setting by greater than 5°F. The thermostatic switches shall conform to MIL-C-2212, and be wired separately from the over-temperature safety device and the low level cutout switch.

3.7.10.2 Controls. A magnetic contactor or relay operating in the control circuit shall be furnished by the heater manufacturer with each water heater. The contactor or relay switch shall be mounted in a corrosion-resistant steel enclosure in accordance with MIL-E-2036 and shall have a removable, corrosion-resistant steel cover for access to the control devices. In addition, the enclosure location shall be in an area to prevent heat from the tank to exceed the operating temperature limits of the control devices. Power leads shall not enter the enclosure through the cover plate. Nuts, bolts, and screws shall be copper-nickel or corrosion-resistant steel. The enclosure gasket shall be made of rubber.

Contactors shall operate on 440 volts, 3 phase, 60 cycle for ac heaters and shall be in compliance with MIL-C-2212. Relays for dc heaters shall be in accordance with MIL-R-2033.

3.7.10.3 Pressure relief valve. Heaters shall have a pressure relief valve with a lifting device constructed and installed in accordance with the ASME BPVC section IV and ANSI Z21.22, with exception to end connections. The relief valve shall be straight threaded union connections fitted with O-rings in accordance with MIL-STD-777, set to relieve at 125 $\pm$ 3 lb/in^2 (see 6.2).

3.7.10.4 Temperature and pressure gauges. Heaters shall have the temperature gauges installed in the inlet and outlet lines. Each heater shall have a pressure gauge located on top of the tank. Each temperature gauge shall have a direct reading 3-inch or larger dial size in accordance with MIL-I-17244. Similarly, each pressure gauge shall have a direct reading 4-1/2 inch or larger dial size in accordance with MIL-G-18997. The temperature gauges shall have a range from 20 to 240°F. Pressure gauge ranges for the supply water line and tank shall be 0 to 200 lb/in^2 gauge in accordance with MIL-G-18997. Thermowells shall be socket welded or socket brazed, 3/4-inch NPS in accordance with MIL-T-24270 and configured to M242701-M2CWR.

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3.7.10.5 Over-temperature safety device. Units shall have an adjustable over-temperature safety device in the outlet water line, in the event that the outlet water temperature rises above 130°F, 160°F, or 180°F temperature setting. The safety device shall trip at the preset temperature (130°F, plus 10°, minus 0°F for class A; 160°F, plus 10°, minus 0°F for class B; and 180°F, plus 10°, minus 0°F for class C) which shall deenergize the heating element(s). The safety device shall simultaneously activate a visual alarm (red telltale light) to alert personnel to an over-temperature condition. The over-temperature safety device shall be wired in series with the low-water cutout switch (see 3.7.10.6). The safety switch shall have a manual reset feature which must be utilized to energize the heating element once a safety shutdown occurs.

3.7.10.6 Low-level cutout device. Units shall have a low-level cutout electrical device installed in the tank in the event of loss or supply water. The low-level cutout electrical device shall be set to deenergize the heating element(s) to prevent overheating. The low-level cutout device shall simultaneously activate a visual alarm (red telltale light) to alert personnel of loss of water supply. The low-level cutout switch shall be wired in series with the over-temperature safety device (see 3.7.10.5). In addition, the device shall have a manual reset feature which must be utilized to energize the heating element(s) once a shutdown occurs.

3.7.10.7 Wiring. Wire used to connect the electrical components shall be in accordance with MIL-M-16878, MIL-M-16878/1, MIL-M-16878/2, MIL-M-16878/3, MIL-M-16878/17, MIL-M-16878/18, and MIL-M-16878/19. Internal wiring shall be completed to an approval terminal enclosure for interconnection of each component for field installation. Wire shall be sized to suit load requirements. Wires more than 12 inches in length shall be marked for identification. Marking shall be in accordance with MIL-E-917.

3.7.10.8 Insulation. Heaters shall be thermal insulated and lagged in accordance with MIL-STD-769 (see 3.7.20). Insulation and lagging shall not interfere with removal of the heater elements or thermostatic switch from the tank.

3.7.10.9 Electrical equipment. Electrical equipment, such as control lights, and so forth, shall operate on 120V, 1-ph, 60-Hz electrical service and conform to applicable requirements of MIL-E-917. DC water heaters shall use dc electrical accessories. For dc water heaters with ac accessories, a warning label plate in accordance with MIL-C-2212 shall be mounted on the front cover of the enclosure to warn service personnel that there are two source of power to the enclosure. Solenoids shall conform to the requirements of MIL-S-4040. Solenoids shall be suitable for continuous duty operation in a 50 degrees Celsius (°C) (122°F) ambient, and shall either have a watertight housing or be encapsulated or both. Electrical equipment shall be suitable for the power supply as stated above unless otherwise specified (see 6.2).

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3.7.10.10 Electromagnetic emission and susceptibility. The monitoring system to enable the system shall meet the requirements of MIL-STD-1399, section 300 and demonstrate compliance in accordance with the general requirements of MIL-STD-461, part 1, and emission and susceptibility requirements and limits CE01, CE03, CE06, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03 of MIL-STD-461, part 5 (see 4.6.11).

3.7.10.11 Electromagnetic pulse (EMP) protection. Monitoring systems produced as specified herein shall survive and operate after being subjected to an EMP environment in accordance with DOD-STD-2169.

3.7.10.12 Electrostatic discharge (ESD). Monitoring systems produced as specified herein shall meet the ESD requirements of MIL-STD-1686, and MIL-STD-454, requirement 75, with the guidance of DOD-HDBK-263.

3.7.10.13 Safety. When specified (see 6.2), a preliminary hazard analysis shall be performed (see 6.3).

3.7.11 Wattage. The wattage rating shall be selected to give the energy required to heat the tank as specified in 1.2 within 1 hour with rated voltage applied to the elements when the tank is filled with water at 40°F. Watt density shall be not greater than 50 watts per square inch of element heating surface.

3.7.12 Dielectric tests. Heaters shall satisfactorily withstand the dielectric strength tests as specified in 4.6.7.

3.7.13 Serviceability. Components in this system shall be free of obstruction when requiring servicing or replacement.

3.7.13.1 Special tools. Special tools required for servicing the heater and associated valves and components shall be furnished by the contractor (see 6.2).

3.7.14 Facilities. Standard shipboard and shore intermediate maintenance facilities capable of plumbing system repair and testing will be used.

3.7.15 Support and test equipment. Built-in test equipment shall be included in the system design as specified in 3.7.10.4.

3.7.16 Painting. A primer pretreatment coating, in accordance with DOD-P-15328, followed by one coat of Navy grey enamel, in accordance with MIL-E-15090, shall be applied to all external services of the heater. Controller shall be painted in accordance with MIL-E-917.

3.7.17 Hydrostatic testing. Components shall withstand an internal hydrostatic test pressure of 250 lb/in^2 (see 4.8.3).

3.7.18 Maintainability demonstration. The first production unit shall be inspected after testing and the capability to maintain, disassemble, and repair the unit shall be demonstrated in accordance with all technical manual requirements (see 4.6.5).

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3.7.19 Shock and vibration. Heater systems shall meet shock and vibration requirements of grade B, deck mounted, class I, type A of MIL-S-901 and type I of MIL-STD-167-1, respectively. The tests shall be conducted at the heater operating temperatures and pressures. The tested system shall include all components required to make a complete and operable heater system, as shown on figure 1.

3.7.20 Surface temperature. Heater components and piping shall be insulated to limit the surface temperature to 125°F maximum when subjected to maximum temperature possible during operation. Insulation shall be in accordance with MIL-STD-769 (see 3.7.10.8).

3.8 Performance requirements.

3.8.1 Full-power test. A full-power test shall be conducted on the water heater systems to ensure proper outlet water temperatures within the response time specified in 3.3.3).

3.8.2 Steady-state performance. Water heater systems shall satisfactorily maintain steady-state performance at all conditions of operation (see 3.5).

3.9 Volume and weight. Components defined in the system envelopes on figure 1 shall be included within the volume and weight constraints specified in table III.

TABLE III. Electric water heater volumes and weights.

Class	AS/BS/CS	A10/B10/C10	A25/B25/C25	A50/B50/C50
Volume L x W x H (inches)	35 x 17 x 26	35 x 18 x 49	38 x 29 x 51	40 x 20 x 51
System dry weight (pounds)	75	170	218	290

3.10 Cleaning kits. When specified (see 6.2), a cleaning kit to remove scale buildup due to calcium deposits shall be furnished.

3.11 Workmanship. Heaters shall be free of visual and operating defects such as: sharp edges, dents, scratches, tears, paint runs, and malfunctioning controls. Mating surfaces of flanges and mounting pads of supports shall be machined true.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.7).

4.3 Instrumentation calibration. Measuring and test equipment used in the performance of tests shall be calibrated in accordance with MIL-STD-45662 (see 6.3).

4.4 First article inspection. Unless otherwise specified (see 6.2), the first heater of each class and capacity shall be subjected to first article inspection which shall consist of the examinations and tests as specified in table IV.

TABLE IV. First article tests.

Tests	Requirement	Test method
Sizes	1.2	
Full power	3.12.1	4.6.4.1
Over-temperature safety device	3.7.10.5	4.6.4.3
Low-level cutout device	3.7.10.6	4.6.4.2
Shock	3.7.19	4.6.1
Vibration	3.7.19	4.6.2
Heater inclination	3.7.8	4.6.3

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TABLE IV. First article tests - Continued.

Tests	Requirement	Test method
Maintainability demonstration	3.7.18	4.6.3
Dielectric tests	3.7.12	4.6.7
Thermostat test	3.7.10.1	4.6.8
Voltage temperature test	3.7.10	4.6.9
Steady-state performance test	3.12.2	4.6.10
Hydrostatic test	3.7.17	4.8.3

4.5 First article testing. Whereas some tests may be able to be conducted concurrently, the shock test, inclined operation test, and vibration test shall be performed separately.

4.5.1 Test failure. Failure of the water heater system to pass any of the tests or examinations specified herein shall constitute rejection of the water heater design.

4.6 First article inspection test method.

4.6.1 Shock test. Hydrostatic testing shall be completed before the shock test is performed (see 4.8.3). The shock test shall be as specified in 3.7.19. Upon completion of the test, the heater system shall be tested as specified in 4.8 through 4.8.3.1.

4.6.1.1 Acceptance criteria. The water heater shall be considered acceptable for shock application unless the following conditions occurred during the test or were evident at the conclusion of the test:

(a) Fragmentation or missile effects of parts.
(b) Displacement or dislodgement of any piece, part or component in such a manner that would be hazardous to personnel in the vicinity of the water heater system or cause an electrical short circuit.
(c) Shearing or tearing of metallic parts that would create projections or sharp edges.
(d) Any potable water (hot or cold) leakage.
(e) Failure to operate as specified in 1.2, 3.7.10.1, 3.7.10.5, 3.7.10.6, and 4.6.7.
(f) Failure to meet requirements as specified in 4.8 through 4.8.3.1.
(g) Post shock tear down inspection reveals internal damage. Post shock tear down inspection can be conducted as part of the maintainability demonstration (see 4.6.5).

Momentary opening and closing of contactor and relay contacts under shock is permitted when operation of the heater is not effected.

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4.6.2 Vibration test. The environmental vibration test shall be conducted in accordance with MIL-STD-167-1 for type I equipment.

4.6.2.1 Acceptance criteria. Acceptance of the water heater system shall be in accordance with MIL-STD-167-1.

4.6.3 Heater inclination. Water heaters as specified in 3.7.8 shall be constructed to operate properly under permanent 15-degree inclination in any direction and symmetrical rolling to 45-degree inclination in any direction for at least 1 hour. The established performance characteristics shall be monitored at 10-minute intervals throughout the test.

4.6.3.1 Acceptance criteria. The water heater system shall be acceptable if the required temperatures do not fluctuate beyond the limits specified in table I and no leakage of fluids occur. Any evidence of failure to the above conditions shall be cause for rejection.

4.6.4 Performance tests.

4.6.4.1 Full-power test. Each water heater element shall be bench tested at full power to ensure the required outlet water temperatures are specified in 3.3.3 (see 6.3). The heating element shall be tested with 40°F water.

4.6.4.1.1 Acceptance criteria. Failure of the water heater element to comply with 3.3.3 shall be cause for rejection.

4.6.4.2 Low-level cutout test. The low-level cutout device shall be tested for each sample water heater system. The heater element shall be energized, the inlet water valve secured and the heater drain valve shall be opened to drain the water. The heater low-level cutout device shall open the circuit and deenergize the heating element when the device is not completely submerged in the water. A red telltale light shall illuminate to indicate deenergization of the heating element has occurred.

4.6.4.2.1 Acceptance criteria. Failure of the water heater system to be as specified in 3.7.10.6 shall be cause for rejection.

4.6.4.3 Over-temperature safety device test. For sampling, heater systems shall have their thermostats set for 130°F for class A, 160°F for class B, and 180°F for class C as specified in 1.2. The over-temperature safety device shall be set as specified in 3.7.10.5. The test shall consist of increasing the outlet water temperature of each class of heater at a rate of at least 1 degree per minute. The safety device shall deenergize the heating element(s) when the specified over-temperature settings are exceeded. The inlet water temperature shall be monitored within 12 inches of the heater water inlet. The outlet water temperatures shall be monitored not more than 12 inches downstream of the outlet water tank fittings. Outlet water temperature readings shall be taken every 15 seconds during the test. A red telltale light shall illuminate to indicate deenergization of the heating element has occurred.

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4.6.4.3.1 Acceptance criteria. Failure of the safety device to deenergize the heating element as specified herein shall be cause for rejection.

4.6.5 Maintainability demonstration. The unit shall be disassembled, all components reassembled and replaced. The controls and instruments shall be removed and replaced. The demonstration shall show that access and replacement of components is possible without personnel hazard and in accordance with procedures outlined in the technical manual (see 3.7.18). The personnel and specific types of tools used in the demonstration shall be identified. In addition, the demonstration shall verify that the MTTR requirement as specified in 3.4 is satisfied by the design and construction of the heater assembly. The demonstration shall be conducted utilizing the instruction manual and the recommended tools without the use of an expert mechanic.

4.6.5.1 Disassembly and reassembly procedure. The disassembly and reassembly procedure of the electric water heater shall include, but not be restricted to, the following items:

(a) Thermostatic switch/well (thermostat).
(b) Heating element.
(c) Contactors and relays.
(d) Over-temperature sensor.
(e) Relief valve.
(f) Low-level cutout device.

4.6.5.2 Acceptance criteria. Failure of the system to be disassembled and reassembled, without personnel hazard, in accordance with procedures outlined in the technical manual, shall be cause for rejection.

4.6.6 Calibration. Calibration of the thermostatic switch shall be checked by a controlled-temperature liquid-bath system, or by thermocouples attached to the sensing bulb of the temperature control to indicate the temperature at which the switch contact open at various settings within the specified range.

4.6.6.1 Acceptance criteria. Failure of the switch contacts to open, at various settings within the specified range, shall be cause for rejection.

4.6.7 Dielectric tests.

4.6.7.1 General. The dielectric tests shall be made after all other first article tests have been completed. The test voltage shall be equal to twice the rated value of the terminal voltage plus 1000 volts. The frequency shall be not less than 60 cycles, and the wave shape shall approximate a sine wave. The test voltage shall be applied continuously for a period of not less than 1 minute.

4.6.7.2 Measurement of test voltage. The measurement of the voltage used in dielectric tests shall be made by means of a voltmeter whereby the meter derives its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer or through an auxiliary potential transformer.

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4.6.7.3 Points of application. The test voltage shall be successfully applied between each electric circuit and the frame with all other electric circuits and metal parts grounded.

4.6.8 Thermostat test. The heater shall be filled with water at approximately 40°F, the temperature taken, and the thermostatic switch set to cut off at approximately 130°F for class A, 160°F for class B, or 180°F for class C heaters. Rated voltage shall be applied to the heating circuit, and the water heated until the thermostatic switch automatically cuts off (see 6.3).

4.6.8.1 Acceptance criteria. Failure of the thermostat to automatically cut off at the present temperature shall be cause for rejection.

4.6.9 Voltage temperature test. The control equipment shall be subjected to a 4-hour operation at the voltage specified to determine that the temperature rise of the current-carrying parts does not exceed the maximum values permitted in accordance with MIL-C-2212.

4.6.9.1 Acceptance criteria. Failure of the current-carrying parts to be in accordance with MIL-C-2212 shall be cause for rejection.

4.6.10 Steady-state performance test. For sampling, sample water heater systems shall have a steady-state performance test (see 6.3). The test shall consist of the monitoring of the heater system's temperature gauges, to determine the temperature swing, while the system is operating as specified in 1.2 and 3.7.10.1. The temperatures shall be monitored for a period of 1 hour with readings taken at 5-minute intervals. After 1 hour, approximately 25 percent of the delivered water shall be drawn from the tank. Temperature readings shall be monitored again for a period of 1 hour at 5-minute intervals. No adjustments shall be made to the heater during the steady-state performance tests other than as specified herein.

4.6.10.1 Acceptance criteria. Failure of the heater system to maintain a temperature swing as specified in 3.7.10.1 shall be cause for rejection.

4.6.11 Electromagnetic interference. Compliance with the requirements of 3.7.10.10 shall be demonstrated by performing and passing electromagnetic emission and susceptibility tests in accordance with MIL-STD-462 (see 6.3).

4.7 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in 4.7.3 and 4.8 (see 6.3).

4.7.1 Lot. Heaters of the same class and size offered for delivery within a time span not greater than 1 year shall be considered a lot.

4.7.2 Sampling for quality conformance. A random sample of heaters shall be selected from each lot as specified in table V for the examination of 4.7.3 (see 6.10).

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TABLE V. Sampling for quality conformance.

Lot size (number of units)	Sample size (number of units)
1 to 6	All
7 to 15	7
16 to 25	10
26 to 40	13
41 to 65	17

4.7.3 Heater system examination. Each heater system in the sample shall be examined and measured to verify compliance with the requirements of materials, workmanship and design of this specification, not involving tests. Each heater system in the sample containing one or more defects shall not be offered for delivery (see 6.10).

4.7.3.1 Additional examinations. Each sample unit selected as specified in 4.7.2 shall be examined for adjustment, fits, leaks, material, finish, and general conformance to this specification and as follows:

(a) The external fittings shall be properly secured.
(b) Bolts, nuts, and screws shall be tight; equipment and accessories shall be properly fastened and secured.
(c) No parts shall be fractured, split, torn, dented, or otherwise damaged to adversely effect serviceability.
(d) There shall be no sharp or ragged edges on any surfaces that may be injurious to personnel.
(e) The temperature control shall be properly set and functioning.

4.8 Quality conformance inspection test methods. The tests specified in 4.8.1 through 4.8.1.1 and table VI shall be conducted on each sample heater of the lot manufactured.

TABLE VI. Quality conformance examination and tests.

Examination and tests	Requirements	Test method
Visual and dimensional	See table VII	4.8.1
Preparation for hydrostatic tests		4.8.2
Hydrostatic testing	3.7.17	4.8.3

4.8.1 Visual and dimensional examinations. Each heater system shall be visually examined to determine conformance with the requirements of table VII.

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TABLE VII. Visual and dimensional inspection.

Inspection	Requirement
Classification	1.2
Sizes	1.2
Materials	3.2
Voltages	3.5
Temperature adjustments	3.6
Tanks	3.7.1
Shells	3.7.1.1
Heads or cover plates	3.7.1.2
Jackets	3.7.2
Flanges	3.7.3
Heating elements	3.7.4
End connections	3.7.5
Welding and silver brazing	3.7.6
Heater supports	3.7.7
Identification and information plates	3.7.9
Thermostatic switches	3.7.10.1
Controls	3.7.10.2
Pressure relief valve	3.7.10.3
Temperature and pressure gauges	3.7.10.4
Over-temperature safety device	3.7.10.5
Low-level cutout device	3.7.10.6
Wiring	3.7.10.7
Insulation	3.7.10.8
Special tools	3.7.13.1
Painting	3.7.16
Workmanship	3.11
Integration requirements	6.9
Dimensions (drawings)	Appendix A

4.8.1.1 Acceptance criteria. Any defects found during inspection as outlined in table VII shall be cause for rejection.

4.8.2 Preparation for hydrostatic testing. The heater system shall be connected for testing as required to simulate shipboard operation. Gauges, thermometers, and controls shall be calibrated before the test to manufacturer specifications and traceable to the National Institute of Standards and Technology (see 4.3). Each heater lot order shall be tested as specified in 4.8.3 to ensure all units are manufactured to the specified requirements.

4.8.3 Hydrostatic testing. Heat transfer elements including vessels, shells, heads, cover plates, and end connections shall be hydrostatically tested to 250 lb/in^2 in accordance with ASME BPVC Section VIII. Temperature and pressure

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gauges shall be isolated during the test. The pressure shall be isolated and maintained in these units for a minimum of 30 minutes. Hydrostatic testing shall be completed before the shock test is performed (see 4.6.1).

4.8.3.1 Acceptance criteria. The failure to maintain 250 lb/in^2 in a unit for 30 minutes, resulting from visible evidence of leakage, permanent distortion, or collapse, shall be cause to reject the unit.

4.9 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packing requirements. Heaters, accessories, and technical data shall be preserved level A, C, or commercial; packed level A, B, C, or commercial as specified (see 6.2); marked in accordance with MIL-E-17555; and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

(a) Navy shipboard stowage fire-retardant requirements.

(1) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, nonweather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of ASTM D 4727.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Storage type, quick recovery, electric water heater systems are to be used onboard ships where continuous hot water generation is required.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Class and size of heater (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) First article inspection (see 3.1).
(e) Materials if other than as specified (see 3.2).
(f) Power supply required (see 3.5).
(g) Inlet water temperature if other than as specified (see 3.5.1).
(h) Inlet water pressure if other than as specified (see 3.5.1).
(i) Withstanding conditions if other than as specified and capacity required under those conditions (see 3.5.2); in particular, voltage ranges that fall outside those specified in 3.5.2(c).
(j) When shell working pressure is other than as specified (see 3.7).
(k) Thermostatic switch adjustment range if other than as specified (see 3.7.10.1).
(l) Shut-off pressure of the inlet water pumps (see 3.7.10.3).
(m) Electrical equipment power requirement if other than as specified (see 3.7.10.9).
(n) When a preliminary hazard analysis is required (see 3.7.10.13).
(o) Special tools if required (see 3.7.13.1).
(p) When cleaning kits are required (see 3.10).
(q) When first article sample is other than as specified (see 4.4).
(r) Level of preservation and packing, and marking required (see 5.1).
(s) Bar codes and packaging acquisition options (see 5.1).
(t) When Navy fire-retardant requirements are other than as specified (see 5.1).
(u) Inclusions to installation instructions (see 6.4.1.1).
(v) Onboard repair parts required if other than as specified (see 6.5.2).
(w) When audio/visual training tape is required (see 6.8).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific

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acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.3	DI-ILSS-81092	Audiovisual aids, master reproducibles and review copies for training equipment and training courses	----
3.3 and appendix A	DI-DRPR-80651	Engineering drawings	----
3.4	DI-RELI-80250	Reliability test plan	----
3.4	DI-RELI-80251	Reliability test procedures	----
3.4	DI-RELI-80252	Reliability test reports	----
3.6, 4.3, 4.4, 4.6.4.1, 4.6.8, 4.6.10, 4.7, and appendix B	DI-NDTI-80809	Test/inspection reports	----
3.6, 4.3, 4.4, 4.6.4.1, 4.6.8, 4.6.10, 4.7, and appendix B	DI-MISC-80678	Certification/data sheet	10.3.1 and 10.3.2 apply
3.7.10.13	DI-SAFT-80101	System safety hazard analysis report	Task 202
4.2	DI-QCIC-81110	Inspection and test plan	----
4.6.5	DI-MNTY-80144	Maintainability demonstration report	----
4.6.11	DI-EMCS-80200	Electromagnetic interference test report	----
4.7	UDI-R-21374	Quality assurance program plan	----
4.7	DI-M-30414	Quality program plan	----
4.7	DI-RELI-80322	Quality conformance inspection and test procedures	----
	DI-NDTI-80492	Production test report	----

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 General. Technical manuals must be in accordance with MIL-M-15071.

6.4.1.1 Descriptions of procedures. Technical manuals should include descriptions of the proper installation of the heater system, installation of temperature gauges, proper electrical connections, and any other procedure to make a safe and operable heater system (see 6.2).

6.4.1.2 Preventive and corrective maintenance procedures. A preventive and corrective maintenance procedure for the heater systems should be described in the technical manuals.

6.4.1.3 Visual maintenance aides. Inclusion of a maintenance index page (MIP) or maintenance requirement cards (MRCs), or both, in accordance with MIL-P-24534 and OPNAVINST 4790.4, should be considered.

6.4.1.4 Packing and quantity of technical manuals. Two copies of the current operation/maintenance technical manual should be packed with each system or equipment. Technical manual packing should be as specified (see 5.1).

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5.1 Onboard repair parts. Unless otherwise specified (see 6.2), onboard repair parts should be furnished for each size heater installed, as specified in table VIII.

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TABLE VIII. Onboard repair parts.

Name of part	Quantity based on number of heaters installed		
	1-5	6-10	11-20
Heater element assy: Thermostatic switch assy: Gaskets and seals:	1 1 100 percent	1 2 100 percent	2 4 100 percent
Contactor (in accordance with MIL-C-2212 or MIL-R-2033):	As applicable		

Complete components and all serviceable parts including onboard repair parts should be listed in accordance with the contract.

6.6 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.7.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Audiovisual training. When specified (see 6.2), an audiovisual training tape should be prepared. The training tape should be a 1-inch video master which illustrates the operation and function, and demonstrates maintenance and repair of the units. The master tape should illustrate the operation, function, maintenance, and repair of both class A, class B, and class C heaters.

6.9 Integration requirements. Water heaters should be complete, compatible systems delivered to the shipyard either preassembled or ready for field assembly except for recirculation systems.

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6.10 Lot acceptance and rejection criteria. If the number of defective systems in any sample exceeds all acceptance numbers for that sample (see table IX), the lot represented by the sample should not be offered for delivery (see 4.7.1).

TABLE IX. Acceptance/rejection numbers.

Lot size (number of units)	Sample size (number of units)	Defective units	
		Acceptance number	Rejection number
1 to 6	All	0	0
7 to 15	7	0	1
16 to 25	10	0	1
26 to 40	13	0	1
41 to 65	17	1	2

6.11 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.12 Part or identifying number (PIN). PINs are composed in the following manner for the electric water heaters covered in this specification:

Example of PIN: MS965-A5

MS 965 - A5

- A5 — Class, size designation (see table I)
- 965 — Specification number
- MS — MS prefix - an item defined by inch-pound units

6.13 Subject term (key word) listing.

Indicator, pressure
Piping, brazed
Solenoids, electrical
Temperature gauges
Thermostat

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6.14 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4520-N003)

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included on the drawings when specified in the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENTS

30.1 Level 2 drawings. Level 2 heater and heater component drawings shall permit evaluation of the design, materials, and construction.

30.2 Level 3 drawings. Level 3 drawings shall include detail manufacturing parts, subassembly, and final assembly drawings, and manufacturing and assembly procedures.

30.3 Limited rights legend. When the Government has only limited rights to the data shown on the drawings, as determined by the contractual provision regarding rights to technical data, the drawings furnished shall be marked with the following restrictive legend:

> "Furnished under United States Government Contract No. _____. Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repairs or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work: or (b) release to a foreign government, as the interests of the United States may require; provided that in either case the release, use, duplication or disclosure hereof shall be subject to the foregoing limitations. This legend shall be marked on any reproduction hereof in whole or in part."

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APPENDIX B

CERTIFICATION/DATA REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included in the certification/data report when specified in the contract or order. This appendix is mandatory only when data item descriptio DI-MISC-80678 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. CERTIFICATION CONTENT

30.1 Instrumentation calibration and recalibration. Original settings, out of tolerance conditions, and retests shall be prepared and certified by the contractor and shall be subject to review and approval by the contracting activity.

30.2 Certification and technical data on passing specific tests. The contractor shall provide certification that the water heater units or their component parts have been tested as specified herein, have passed the test, and are in compliance with this specification. Test data upon which the certification is based shall be subject to review and approval by the contracting activity for these tests:

(a) First article inspection (see 4.4).
(b) Full-power test (see 4.6.4.1).
(c) Thermostat test (see 4.6.8).
(d) Steady-state performance test (see 4.6.10).
(e) Quality conformance inspection (see 4.7).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-965E(SH)	94/12/22

3. DOCUMENT TITLE

HEATERS, WATER, ELECTRIC, NAVAL SHIPBOARD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY TECHNICAL POINT OF CONTACT:

a. NAME	b. TELEPHONE (Include Area Code)	
Ted Halverson 03Y2	(1) Commercial	(2) AUTOVON
PLEASE SEND ALL CORRESPONDENSE TO:	(703) 602-0367	332-0367
ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 03R42, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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